QUICK START & SAFETY GUIDE · SKU CQA240229JN

From a Bare Jetson Nano to Live EEG

Eight steps: prepare the Jetson Nano, enable SPI, mount the shield, connect the electrodes, power on from the battery, start PiEEG Server and check the signal.



1 – JNEEG, 2 – Clips Electrodes, 3 – EEG cap with, 4 – power battery

WARNING Battery only.

JNEEG has **no isolation circuitry**. The Jetson, the shield and everything cabled to them (monitor, keyboard, mouse) must run from a **5 V power bank** while electrodes are on a person. JNEEG is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only.** Jetson, shield and everything cabled to them run from a 5 V power bank.
2	**No mains connection of any kind:** not the Jetson's barrel-jack supply, no wall charger, no HDMI/USB to a mains-powered monitor or computer, no Ethernet cable.
3	**No diagnosis.** Do not use JNEEG to diagnose, monitor or treat any medical condition.
4	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
5	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off gel after each session.
6	**Exposed electronics.** No case and the Jetson heatsink gets hot: handle by the edges, keep liquids away.

Power while recording

Setup	While electrodes are on	Why
Jetson + JNEEG on a 5 V power bank	SAFE	No path to the mains.
Monitor powered by the bank or the Jetson	SAFE	Still fully on battery.
Laptop on battery, dashboard over Wi-Fi	SAFE	No cable to the Jetson or the person.
Jetson on its wall power supply	NEVER	Connects the person to the mains through the electrodes.
Monitor plugged into a wall socket	NEVER	The HDMI cable links the mains to the Jetson.
Ethernet / USB to a mains-powered device	NEVER	Same reason.

02 WHAT YOU NEED

Item	Requirement
JNEEG	Included
Jetson Nano Developer Kit	microSD 32 GB+ with NVIDIA JetPack
5 V power bank	5 V, at least 3 A, up to 10,000 mAh; short, thick USB cable
Electrodes	8 dry or wet Ag/AgCl with 2.54 mm connectors + 2 ear clips
EEG cap	10-20 cap
Monitor, keyboard, mouse	Monitor powered from the bank or the Jetson; wired keyboard and mouse — or a battery-powered laptop for headless use
Gel	Only for wet electrodes

03 SETUP IN EIGHT STEPS

1 Gather the parts

JNEEG, Jetson Nano, charged power bank, 8 electrodes and 2 ear clips, cap.

2 Prepare the Jetson Nano

Flash NVIDIA JetPack for the Jetson Nano Developer Kit to the microSD card, finish first-run setup and connect to the internet for installing software.

3 Enable SPI

TERMINAL ON THE JETSON

```
sudo /opt/nvidia/jetson-io/jetson-io.py   # Configure 40-pin header -> spi1 -> Save and reboot
ls /dev/spidev0.0
sudo groupadd -f spi && sudo usermod -aG spi,gpio "$USER"   # then log out and in
```

4 Mount the shield

Shut down the Jetson and unplug it. Line up pin 1 with pin 1 on the 40-pin header and press down evenly until every pin is seated.

5 Connect the electrodes

Channels 1–8 to the cap, **Ref** to one earlobe, **Biasout** to the other (JN-PIN-004).

6 Power on from the battery

Connect the battery-powered monitor and wired keyboard/mouse, then plug the power bank into the Jetson's Micro-USB input. LED D1 on the shield lights.

7 Install and start PiEEG Server

NEEDS PYTHON 3.10+

```
python3 -m venv .venv && source .venv/bin/activate
pip install --upgrade pip && pip install "pieeg-server[rpi]"
pieeg-server doctor
pieeg-server --device jneeg
```

JetPack 4 ships Python 3.6: install a newer Python first, or use the Python scripts (JN-SW-006), which run on the stock Python.

8 Check the signal

Open **http://localhost:1617** on the Jetson (or http://<jetson-ip>:1617 from a battery-powered laptop) and click **Connect**.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All 8 channels settle. Flat lines or huge values mean a loose electrode.
Close the eyes for 10 s	Alpha (8–13 Hz) rises on O1 and O2, then drops when the eyes open.
Blink a few times	Large slow spikes on Fp1 and Fp2.
Clench the jaw	Bursts of high-frequency noise — muscle (EMG), not brain activity.

05 TROUBLESHOOTING

Power and hardware

Problem	Fix
LED D1 on the shield is off	Check the power bank is on and the shield is seated on all 40 pins.
Jetson shuts down or restarts	Bank too weak: use 5 V / 3 A and a short, thick cable. Use the 5 W mode (`sudo nvpmodel -m 1`).
Shield off by one pin	Shut down, remove and re-seat; pin 1 must match pin 1.

SPI, GPIO and software

Problem	Fix
`/dev/spidev0.0` not found	Enable spi1 with jetson-io.py and reboot.
Permission denied on SPI/GPIO	`sudo usermod -aG spi,gpio $USER`, then log in again.
PiEEG Server: no samples	DRDY line offset may differ: check `gpioinfo gpiochip0`, pass `--gpio-chip /dev/gpiochip0`.
pip cannot install pieeg-server	It needs Python 3.10+; JetPack 4 ships 3.6.
Dashboard shows 16 channels	Start with --device jneeg.
Jetson.GPIO errors in scripts	Run as a user in the gpio group, or with sudo.

Signal

Problem	Fix
All channels noisy at once	Check the **REF and BIAS ear clips** first.
One channel flat or very large	That electrode is not touching the skin.
Strong 50/60 Hz hum	Something is on the mains — unplug every charger, run the monitor from the battery.
Noise from the fan	Keep electrode cables away from the Jetson fan and heatsink.
Slow spikes on front channels	Blinks and eye movements.
Bursts of high-frequency noise	Jaw, face or neck muscles.

> **NOTE Board or setup?**
> Set all channels to input-shorted (`CHnSET = 0x01`). If the signal stays within a few µV, the hardware is fine and the noise comes from cables, electrodes or surroundings.

06 RELATED DOCUMENTS

JN-DS-001 Product Datasheet	**JN-MD-002** Mechanical & Dimensional Drawing	**JN-PCB-003** PCB & Manufacturing Specification
JN-PIN-004 Connector Pinout & Electrode Guide	**JN-SW-006** Data Interface & Software Guide	

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